

UNALAKLEET, AK, USA

WMO: 702070

Lat: 63.883N

Lon: 160.800W

Elev: 18

StdP: 14.69

Time Zone: -9.00 (NAK)

Period: 94-19

WBAN: 26627

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-29.3	-23.6	-29.2	1.1	-19.9	-27.4	1.2	-17.0	42.8	12.2	39.2	12.9	3.8	130	0.869

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.9	69.8	58.3	65.6	57.3	63.2	55.3	60.2	66.6	58.2	64.3	56.4	62.0	11.1	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
56.8	68.8	63.3	54.7	63.7	61.3	53.6	61.2	60.2	26.5	66.9	25.1	64.2	23.9	61.9	68.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 34.3	30.2	26.7	DB	-32.8	78.3	8.8	4.4	-39.1	81.4	-44.3	84.0	-49.3	86.5	-55.7	89.6	(4)
(5)			WB	-28.8	62.3	5.4	3.1	-32.7	64.5	-35.9	66.3	-38.9	68.0	-42.9	70.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	29.6	7.2	13.0	11.7	26.2	39.4	50.0	55.5	52.5	45.3	29.4	13.9	10.4		
	DBStd	20.86	17.59	14.32	13.86	11.74	8.47	6.44	4.65	5.29	6.27	10.41	12.99	14.28		
	HDD50	7826	1327	1036	1188	714	337	75	7	32	161	638	1082	1229		
	HDD65	12912	1792	1456	1653	1164	794	450	295	388	591	1103	1532	1694		
	CDD50	393	0	0	0	0	9	76	179	109	20	0	0	0		
	CDD65	4	0	0	0	0	0	1	2	1	0	0	0	0		
	CDH74	33	0	0	0	0	1	16	13	3	0	0	0	0		
	CDH80	4	0	0	0	0	0	3	1	0	0	0	0	0		
	WSAvg	11.4	14.7	15.4	11.6	10.5	8.5	8.4	9.4	9.8	10.1	11.0	13.4	14.4		
	PrecAvg	12.9	0.5	0.4	0.5	0.6	0.7	1.3	2.2	2.9	2.2	1.0	0.7	0.7		
(15) Precipitation	PrecMax	19.7	2.0	1.2	2.2	2.0	2.5	5.0	4.4	7.4	6.6	2.9	2.0	2.0		
	PrecMin	8.0	0.0	0.0	0.0	0.0	0.0	0.2	0.0	1.1	0.4	0.2	0.0	0.0		
	PrecStd	3.6	0.4	0.4	0.5	0.6	0.6	1.1	1.3	1.5	1.7	0.6	0.5	0.5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.2	39.1	40.7	52.3	66.2	75.1	75.1	72.1	63.8	53.5	39.0	38.5		
		MCWB	35.2	34.4	35.0	41.4	50.7	56.6	59.9	60.2	52.5	46.0	35.1	34.6		
	2%	DB	36.5	36.4	36.8	47.7	58.4	66.8	69.9	66.3	58.9	47.3	36.5	35.9		
		MCWB	32.8	32.8	32.4	39.3	47.5	56.3	59.4	57.9	50.3	42.7	33.2	32.8		
	5%	DB	33.6	34.3	34.2	43.6	54.3	63.2	65.9	63.6	55.5	45.1	34.3	32.0		
		MCWB	29.8	31.2	30.5	36.8	46.0	53.9	58.6	56.1	49.6	40.8	31.5	29.3		
	10%	DB	29.5	30.4	30.2	39.9	51.6	60.6	63.4	61.1	54.2	42.8	30.5	28.0		
		MCWB	27.1	28.0	27.5	34.7	44.3	53.2	56.2	54.9	49.0	38.9	28.5	25.4		
	0.4%	WB	35.5	35.4	35.7	42.2	52.9	60.7	64.0	62.1	54.1	46.3	36.5	35.4		
		MCDB	38.9	38.7	40.1	50.3	64.1	70.1	69.1	68.8	59.5	50.8	38.5	37.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	33.3	33.2	33.0	39.5	49.5	57.3	60.8	59.3	51.9	43.5	33.5	32.6		
		MCDB	36.3	36.2	36.5	46.5	56.6	64.4	66.9	64.7	56.8	46.7	36.3	35.2		
	5%	WB	30.1	31.1	30.5	37.4	46.6	55.2	59.1	57.4	50.5	41.2	31.4	29.3		
		MCDB	32.5	33.9	33.7	43.5	53.3	62.3	65.4	62.8	55.2	44.8	33.9	31.7		
	10%	WB	26.7	28.3	27.6	35.2	44.4	53.2	57.2	55.5	49.2	39.1	28.7	25.8		
		MCDB	28.9	30.7	30.2	40.1	50.0	59.7	63.0	60.0	53.7	42.7	31.0	28.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.5	11.1	13.6	13.2	12.2	11.4	9.9	10.7	10.9	9.9	10.6	10.3		
		MCDBR	10.0	10.3	11.8	14.6	19.2	18.8	16.9	15.4	13.7	9.5	9.2	9.2		
		MCWBR	8.7	8.8	9.5	9.9	11.6	10.6	9.7	9.6	8.9	7.1	8.0	7.8		
		MCDBR	10.4	10.4	10.9	14.0	17.5	16.3	14.4	13.5	11.7	8.4	9.0	8.9		
	5% WB	MCWBR	9.2	9.3	9.2	9.7	11.2	10.1	9.0	8.9	8.2	6.6	8.0	7.8		
(40) Clear-Sky Solar Irradiance	taub		0.194	0.262	0.275	0.337	0.365	0.370	0.366	0.385	0.319	0.284	0.196	0.188		
	taud		1.951	2.155	2.221	2.110	2.255	2.358	2.416	2.360	2.527	2.370	1.964	1.897		
	Ebn at Noon		126	210	261	263	267	269	266	248	248	204	119	50		
	Edh at Noon		9	19	27	39	37	34	31	30	21	16	8	4		
(44) All-Sky Solar Radiation	RadAvg		61	259	818	1391	1594	1686	1398	1079	683	298	85	23		
	RadStd		9	34	78	90	144	163	94	134	69	40	14	2		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page